MPG No. 20.4.74 $\bar{3}'m'$ (-3'm'1 setting) [Type III, trigonal] [M tensor]

* Rank 0 tensor.

$$[M]$$

$$M = \mathbb{M}_0^{(1)}$$

* Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} M_{xx} & 0 & 0 \\ 0 & M_{xx} & 0 \\ 0 & 0 & M_{zz} \end{bmatrix}$$

$$M_{xx} = \mathbb{M}_0^{(1)} - \mathbb{M}_u^{(1)}$$
$$M_{zz} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_u^{(1)}$$

* Rank 2 tensor (a). * Rank 3 tensor (ss). * Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss).

$$\begin{bmatrix} M_{xxxx} & M_{xxyy} & M_{xxzz} & M_{xxyz} & 0 & 0 \\ M_{xxyy} & M_{xxxx} & M_{xxzz} & -M_{xxyz} & 0 & 0 \\ M_{xxzz} & M_{xxzz} & M_{zzzz} & 0 & 0 & 0 \\ M_{xxyz} & -M_{xxyz} & 0 & M_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & M_{yzyz} & M_{xxyz} \\ 0 & 0 & 0 & 0 & M_{xxyz} & \frac{M_{xxxx}}{2} - \frac{M_{xxyy}}{2} \end{bmatrix}$$

$$M_{xxxx} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_0^{(2)} + 3\mathbb{M}_{40}^{(1)} - 2\mathbb{M}_u^{(1)} - 4\mathbb{M}_u^{(2)}$$
$$M_{xxyy} = \mathbb{M}_0^{(1)} + \mathbb{M}_{40}^{(1)} - 2\mathbb{M}_u^{(1)}$$
$$M_{xxzz} = \mathbb{M}_0^{(1)} - 4\mathbb{M}_{40}^{(1)} + \mathbb{M}_u^{(1)}$$
$$M_{xxyz} = \mathbb{M}_{4s}^{(1)}$$
$$M_{zzzz} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_0^{(2)} + 8\mathbb{M}_{40}^{(1)} + 4\mathbb{M}_u^{(1)} + 8\mathbb{M}_u^{(2)}$$
$$M_{yzyz} = \mathbb{M}_0^{(2)} - 4\mathbb{M}_{40}^{(1)} + \mathbb{M}_u^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & M_{xxzz} & M_{xxyz} & 0 & 0 \\ 0 & 0 & M_{xxzz} & -M_{xxyz} & 0 & 0 \\ -M_{xxzz} & -M_{xxzz} & 0 & 0 & 0 & 0 \\ -M_{xxyz} & M_{xxyz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -M_{xxyz} \\ 0 & 0 & 0 & 0 & M_{xxyz} & 0 \end{bmatrix}$$

$$M_{xxzz} = 3\mathbb{M}_u^{(3)}$$
$$M_{xxyz} = 2\mathbb{T}_{3c}^{(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} M_{yzyz} & 0 & 0 \\ 0 & M_{yzyz} & 0 \\ 0 & 0 & M_{xyxy} \end{bmatrix}$$

$$M_{yzyz} = \mathbb{M}_0^{(3)} - 2\mathbb{M}_u^{(4)}$$
$$M_{xyxy} = \mathbb{M}_0^{(3)} + 4\mathbb{M}_u^{(4)}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} M_{xxyz} & 0 & 0 \\ -M_{xxyz} & 0 & 0 \\ 0 & 0 & 0 \\ M_{yzyz} & 0 & 0 \\ 0 & -M_{yzyz} & 0 \\ 0 & -M_{xxyz} & 0 \end{bmatrix}$$

$M_{xxyz} = \mathbb{T}_{3c}^{(2)}$
$M_{yzyz} = -3\mathbb{M}_{u}^{(5)}$

* Rank 4 tensor (as).

$$\begin{bmatrix} M_{yzxx} & -M_{yzxx} & 0 & M_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & -M_{yzyz} & -M_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$M_{yzxx} = \mathbb{T}_{3c}^{(3)}$
$M_{yzyz} = -3\mathbb{M}_{u}^{(6)}$

* Rank 4 tensor (s).

$$\begin{bmatrix} M_{xxxx} & M_{xxyy} & M_{xxzz} & M_{xxyz} & 0 & 0 & M_{xxzy} & 0 & 0 \\ M_{xxyy} & M_{xxxx} & M_{xxzz} & -M_{xxyz} & 0 & 0 & -M_{xxzy} & 0 & 0 \\ M_{zzxx} & M_{zzxx} & M_{zzzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ M_{yzxx} & -M_{yzxx} & 0 & M_{yzyz} & 0 & 0 & M_{yzzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & M_{yzzy} & M_{yzxx} & 0 & M_{yzyz} & M_{yzxx} \\ 0 & 0 & 0 & 0 & M_{xxzy} & \frac{M_{xxxx}}{2} - \frac{M_{xxyy}}{2} & 0 & M_{xxyz} & \frac{M_{xxxx}}{2} - \frac{M_{xxyy}}{2} \end{bmatrix}$$

$M_{xxxx} = \mathbb{M}_{0}^{(1)} + 2\mathbb{M}_{0}^{(2)} + 3\mathbb{M}_{40}^{(1)} - 2\mathbb{M}_{u}^{(1)} - 4\mathbb{M}_{u}^{(2)}$
$M_{xxyy} = \mathbb{M}_{0}^{(1)} + \mathbb{M}_{40}^{(1)} - 2\mathbb{M}_{u}^{(1)}$
$M_{xxzz} = \mathbb{M}_{0}^{(1)} - 4\mathbb{M}_{40}^{(1)} + \mathbb{M}_{u}^{(1)} + 3\mathbb{M}_{u}^{(3)}$
$M_{xxyz} = \mathbb{M}_{4s}^{(1)} + 2\mathbb{T}_{3c}^{(1)} + \mathbb{T}_{3c}^{(2)}$
$M_{xxzy} = \mathbb{M}_{4s}^{(1)} + 2\mathbb{T}_{3c}^{(1)} - \mathbb{T}_{3c}^{(2)}$
$M_{zzxx} = \mathbb{M}_{0}^{(1)} - 4\mathbb{M}_{40}^{(1)} + \mathbb{M}_{u}^{(1)} - 3\mathbb{M}_{u}^{(3)}$
$M_{zzzz} = \mathbb{M}_{0}^{(1)} + 2\mathbb{M}_{0}^{(2)} + 8\mathbb{M}_{40}^{(1)} + 4\mathbb{M}_{u}^{(1)} + 8\mathbb{M}_{u}^{(2)}$
$M_{yzxx} = \mathbb{M}_{4s}^{(1)} - 2\mathbb{T}_{3c}^{(1)}$
$M_{yzyz} = \mathbb{M}_{0}^{(2)} - 4\mathbb{M}_{40}^{(1)} + \mathbb{M}_{u}^{(2)} - 3\mathbb{M}_{u}^{(5)}$
$M_{yzzy} = \mathbb{M}_{0}^{(2)} - 4\mathbb{M}_{40}^{(1)} + \mathbb{M}_{u}^{(2)} + 3\mathbb{M}_{u}^{(5)}$

* Rank 4 tensor (a).

$$\begin{bmatrix} M_{yzxx} & -M_{yzxx} & 0 & M_{yzyz} & 0 & 0 & M_{yzzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & -M_{yzzy} & -M_{yzxx} & 0 & -M_{yzyz} & -M_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & M_{xyxy} & 0 & 0 & -M_{xyxy} \end{bmatrix}$$

$M_{yzxx} = \mathbb{T}_{3c}^{(3)}$
$M_{yzyz} = \mathbb{M}_{0}^{(3)} - 2\mathbb{M}_{u}^{(4)} - 3\mathbb{M}_{u}^{(6)}$
$M_{yzzy} = -\mathbb{M}_{0}^{(3)} + 2\mathbb{M}_{u}^{(4)} - 3\mathbb{M}_{u}^{(6)}$
$M_{xyxy} = \mathbb{M}_{0}^{(3)} + 4\mathbb{M}_{u}^{(4)}$

* Rank 4 tensor (t).

$$\begin{bmatrix} M_{xxxx} & 0 & 0 \\ 0 & M_{xxxx} & M_{yyyz} \\ 0 & 0 & M_{zzzz} \\ 0 & M_{yyyz} & M_{yyzz} \\ M_{zzxx} & 0 & 0 \\ 0 & M_{xxyy} & -M_{yyyz} \\ 0 & M_{yzzy} & 0 \\ 0 & -M_{yyyz} & -\frac{M_{xxxx}}{2} + \frac{3M_{xxyy}}{2} + M_{yyzz} - M_{yzzy} + M_{zzxx} \\ \frac{M_{xxxx}}{2} - \frac{M_{xxyy}}{2} & 0 & 0 \\ -M_{yyyz} & 0 & 0 \end{bmatrix}$$

$$M_{xxxx} = 3\mathbb{M}_0^{(1)} + 3\mathbb{M}_{40}^{(1)} - 2\mathbb{M}_u^{(1)} - 4\mathbb{M}_u^{(2)}$$
$$M_{yyyz} = -\mathbb{M}_{4s}^{(1)}$$
$$M_{zzzz} = 3\mathbb{M}_0^{(1)} + 8\mathbb{M}_{40}^{(1)} + 4\mathbb{M}_u^{(1)} + 8\mathbb{M}_u^{(2)}$$
$$M_{yyzz} = \mathbb{M}_0^{(1)} - 4\mathbb{M}_{40}^{(1)} + \mathbb{M}_u^{(1)} + 3\mathbb{M}_u^{(3)}$$
$$M_{zzxx} = \mathbb{M}_0^{(1)} - 4\mathbb{M}_{40}^{(1)} + \mathbb{M}_u^{(1)} - 3\mathbb{M}_u^{(3)}$$
$$M_{xxyy} = \mathbb{M}_0^{(1)} + \mathbb{M}_{40}^{(1)} - 2\mathbb{M}_u^{(1)}$$
$$M_{yzzy} = \mathbb{M}_0^{(1)} - 4\mathbb{M}_{40}^{(1)} + \mathbb{M}_u^{(2)} + 3\mathbb{M}_u^{(5)}$$